Advanced Statistical Methods II

Lecture 18: Non-linear Regression and Least Squares

1 The general non-linear regression model

The general model has the form

$$y = f(\mathbf{x}; \boldsymbol{\theta}) + \varepsilon,$$

where ε is an error term. The function f is non-linear *in the unknown parameters*. Non-linearity in $\mathbf{x}$ alone does not make a regression model non-linear in this sense.

We observe $(\mathbf{x}_1, y_1), \ldots, (\mathbf{x}_n, y_n)$, where

$$\mathbf{x}_i \in \mathbb{R}^q, \qquad \boldsymbol{\theta} \in \mathbb{R}^p, \qquad f : \mathbb{R}^{q+p} \longrightarrow \mathbb{R}.$$

Given these observations, we wish to estimate $\boldsymbol{\theta}$.

1.1 Assumptions and frameworks

The functional form of f is completely known, its parameter values are unknown. We assume that the explanatory variables are observed without error. An errors-in-variables model would require a different treatment. With a fixed design and a fixed parameter, $f(\mathbf{x}; \boldsymbol{\theta})$ is non-random.

In the *additive framework*,

$$y = f(\mathbf{x}; \boldsymbol{\theta}) + \varepsilon.$$

Two approaches to the parameter are distinguished:

1. Frequentist setup: $\boldsymbol{\theta}$ is fixed but unknown.
2. Bayesian setup: $\boldsymbol{\theta}$ is treated as a random variable.

The following development uses the frequentist setup.

In the *multiplicative framework*,

$$y = f(\mathbf{x}; \boldsymbol{\theta})\varepsilon.$$

When the quantities are positive, logarithms give

$$\log y = \log f(\mathbf{x}; \boldsymbol{\theta}) + \log \varepsilon.$$

Thus a multiplicative error becomes additive on the logarithmic scale.

The main questions of interest are:

1. How do we estimate $\boldsymbol{\theta}$?
2. What properties do these estimators have?
3. How do we find these estimators in practice?
4. Which algorithms are effective in estimating the parameters?
5. How does the particular form of f determine a suitable algorithm?

1.2 Comparison with linear regression

Under the usual full-rank linear-model assumptions, least-squares estimators are unbiased for zero-mean errors. Under Normal errors, they have the usual uniformly minimum-variance unbiased estimation (UMVUE) property. These are finite-sample properties.

Regarding asymptotic properties: a standard design condition for the least squares estimators to be consistent is

$$\frac{1}{n}X^{\mathsf{T}}X \longrightarrow A, \qquad A \text{ positive definite.}$$

For general non-linear regression modeles, we can only discuss the asymptotic properties of the estimators. Exact finite-sample formulas are generally unavailable.

In this lecture, we shall discuss least-squares (LS) estimators.

1.3 Example: a ratio of polynomials

Consider

$$y_t = \frac{a_0 + a_1 t + a_2 t^2 + \cdots + a_p t^p}{b_0 + b_1 t + b_2 t^2 + \cdots + b_q t^q} + \varepsilon_t.$$

Here ε_t is additive, and the errors are i.i.d., with

$$\mathbb{E}(\varepsilon_t) = 0, \qquad \mathrm{Var}(\varepsilon_t) = \sigma^2.$$

The coefficient vector $(a_0, \ldots, a_p, b_0, \ldots, b_q)$ lies in $\mathbb{R}^{p+q+2}$. From the observed pairs (t_i, y_{t_i}), $i = 1, \ldots, n$, we wish to estimate these coefficients.

Remark 1. The denominator must be nonzero at the design points. Multiplying every numerator and denominator coefficient by the same nonzero constant leaves the ratio unchanged. Unique coefficient estimation therefore requires an identifying normalization, such as fixing a nonzero denominator coefficient.

Remark 2 (Optimal Design.). We may choose the n time points $t_1, \ldots, t_n \in [0, T]$ to obtain the "best" possible estimation of the unknown parameters. We compare the resulting variance–covariance matrices of the estimators.

1.4 Example: a sum of exponentials

Another non-linear model is

$$y_t = a_0 + a_1 e^{\beta_1 t} + \cdots + a_p e^{\beta_p t} + \varepsilon_t.$$

It is linear in the a_j when the β_j are fixed, but non-linear in the full set of unknown parameters.

2 Least-squares estimation and the normal equations

Write

$$y_i = f(\mathbf{x}_i; \boldsymbol{\theta}) + \varepsilon_i, \qquad i = 1, \ldots, n,$$

with i.i.d. errors satisfying $\mathbb{E}(\varepsilon_i) = 0$ and $\mathrm{Var}(\varepsilon_i) = \sigma^2$. Assume that f is smooth in $\boldsymbol{\theta}$ and the required derivatives exist.

The parameter space $\Theta \subseteq \mathbb{R}^p$ is taken to be closed and bounded, hence compact. This restriction is used for theoretical purposes. Ordinary unrestricted linear regression typically uses $\Theta = \mathbb{R}^p$.

Define

$$Q(\boldsymbol{\theta}) = \sum_{i=1}^{n} \{y_i - f(\mathbf{x}_i; \boldsymbol{\theta})\}^2.$$

An LS estimator is

$$\widehat{\boldsymbol{\theta}} \in \arg\min_{\boldsymbol{\theta} \in \Theta} Q(\boldsymbol{\theta}).$$

Continuity and compactness ensure that a minimum exists. We assume that the constrained and unrestricted minima agree almost everywhere:

$$\min_{\boldsymbol{\theta} \in \Theta} Q(\boldsymbol{\theta}) = \min_{\boldsymbol{\theta} \in \mathbb{R}^p} Q(\boldsymbol{\theta}).$$

Remark 3. Differentiating and setting derivatives to zero applies to an interior minimizer (or an unrestricted differentiable minimizer). A stationary point need not be a minimum, and boundary minima require separate consideration.

2.1 Coordinate form

For $j = 1, \ldots, p$, we have

$$\begin{aligned}
\frac{\partial Q(\boldsymbol{\theta})}{\partial \theta_j} &= \sum_{i=1}^{n} \frac{\partial}{\partial \theta_j} \{y_i - f(\mathbf{x}_i; \boldsymbol{\theta})\}^2 \\
&= \sum_{i=1}^{n} 2\{y_i - f(\mathbf{x}_i; \boldsymbol{\theta})\} \frac{\partial}{\partial \theta_j} \{y_i - f(\mathbf{x}_i; \boldsymbol{\theta})\} \\
&= -2 \sum_{i=1}^{n} \{y_i - f(\mathbf{x}_i; \boldsymbol{\theta})\} \frac{\partial f(\mathbf{x}_i; \boldsymbol{\theta})}{\partial \theta_j}.
\end{aligned}$$

Equating these derivatives to zero yields the normal equations

$$\sum_{i=1}^{n} \{y_i - f(\mathbf{x}_i; \boldsymbol{\theta})\} \frac{\partial f(\mathbf{x}_i; \boldsymbol{\theta})}{\partial \theta_j} = 0, \qquad j = 1, \ldots, p. \tag{1}$$

2.2 Matrix form

Let

$$\mathbf{y} = \begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix}, \qquad f_i(\boldsymbol{\theta}) = f(\mathbf{x}_i; \boldsymbol{\theta}), \qquad \mathbf{f}(\boldsymbol{\theta}) = \begin{pmatrix} f_1(\boldsymbol{\theta}) \\ \vdots \\ f_n(\boldsymbol{\theta}) \end{pmatrix}.$$

The $n \times p$ Jacobian matrix is

$$J(\boldsymbol{\theta}) = \left(\frac{\partial f_i(\boldsymbol{\theta})}{\partial \theta_j} \right)_{i,j} = \begin{pmatrix} \frac{\partial f_1}{\partial \theta_1} & \cdots & \frac{\partial f_1}{\partial \theta_p} \\ \vdots & \ddots & \vdots \\ \frac{\partial f_n}{\partial \theta_1} & \cdots & \frac{\partial f_n}{\partial \theta_p} \end{pmatrix}.$$

The jth entry of $J(\boldsymbol{\theta})^\mathsf{T}\{\mathbf{y} - \mathbf{f}(\boldsymbol{\theta})\}$ is the sum in (1). Hence

$$\underbrace{J(\boldsymbol{\theta})^\mathsf{T}}_{p \times n} \underbrace{\{\mathbf{y} - \mathbf{f}(\boldsymbol{\theta})\}}_{n \times 1} = \mathbf{0}.$$

In particular, an interior LS solution satisfies

$$J(\widehat{\boldsymbol{\theta}})^\mathsf{T}\{\mathbf{y} - \mathbf{f}(\widehat{\boldsymbol{\theta}})\} = \mathbf{0}.$$

3 Residuals and projection geometry

The residual vector is

$$\widehat{\boldsymbol{\varepsilon}} = \mathbf{y} - \mathbf{f}(\widehat{\boldsymbol{\theta}}).$$

Thus the normal equations say $J(\widehat{\boldsymbol{\theta}})^\mathsf{T}\widehat{\boldsymbol{\varepsilon}} = \mathbf{0}$. If $J(\widehat{\boldsymbol{\theta}})$ has full column rank $p < n$, define

$$P_{J(\widehat{\boldsymbol{\theta}})} = J(\widehat{\boldsymbol{\theta}})\{J(\widehat{\boldsymbol{\theta}})^\mathsf{T} J(\widehat{\boldsymbol{\theta}})\}^{-1} J(\widehat{\boldsymbol{\theta}})^\mathsf{T}.$$

This is the orthogonal projection onto the column space of $J(\widehat{\boldsymbol{\theta}})$. We have

$$J(\widehat{\boldsymbol{\theta}})^\mathsf{T}\widehat{\boldsymbol{\varepsilon}} = \mathbf{0} \quad \Longleftrightarrow \quad P_{J(\widehat{\boldsymbol{\theta}})}\widehat{\boldsymbol{\varepsilon}} = \mathbf{0}.$$

For completeness, let X be any $n \times p$ full-column-rank matrix and $v \in \mathbb{R}^n$. If $X^\mathsf{T} v = 0$, then

$$X(X^\mathsf{T}X)^{-1}X^\mathsf{T}v = 0.$$

Conversely, if $X(X^\mathsf{T}X)^{-1}X^\mathsf{T}v = 0$, premultiplication by X^T gives

$$X^\mathsf{T}X(X^\mathsf{T}X)^{-1}X^\mathsf{T}v = X^\mathsf{T}v = 0.$$

Thus the residual is orthogonal to the local tangent directions of the fitted regression surface. In most non-linear models, the normal equations cannot be solved explicitly: numerical procedures are needed.

4 A simple example: a power-law regression

Consider

$$y_i = \alpha x_i^\beta + \varepsilon_i, \qquad i = 1, \ldots, n,$$

where α, β are unknown, the errors are i.i.d., $\mathbb{E}(\varepsilon_i) = 0$, and $\mathrm{Var}(\varepsilon_i) = \sigma^2$. Take $x_i > 0$ so that arbitrary real powers and $\log x_i$ are defined. The criterion is

$$Q(\alpha, \beta) = \sum_{i=1}^{n} (y_i - \alpha x_i^\beta)^2.$$

4.1 Differentiating with respect to α

$$\begin{aligned}\frac{\partial Q}{\partial \alpha} &= 2\sum_{i=1}^{n}(y_i - \alpha x_i^{\beta})(-x_i^{\beta}) \\ &= -2\sum_{i=1}^{n}(y_i - \alpha x_i^{\beta})x_i^{\beta}.\end{aligned}$$

The first normal equation is therefore

$$\begin{aligned}0 &= \sum_{i=1}^{n}(y_i - \alpha x_i^{\beta})x_i^{\beta} \\ &= \sum_{i=1}^{n} y_i x_i^{\beta} - \alpha \sum_{i=1}^{n} x_i^{2\beta}.\end{aligned}$$

For a fixed β this gives

$$\widehat{\alpha}(\beta) = \frac{\sum_{i=1}^{n} y_i x_i^{\beta}}{\sum_{i=1}^{n} x_i^{2\beta}}. \tag{2}$$

Because

$$\frac{\partial^2 Q}{\partial \alpha^2} = 2\sum_{i=1}^{n} x_i^{2\beta} > 0,$$

this is the unique unrestricted minimizing α for that fixed β.

4.2 Differentiating with respect to β

One has

$$\frac{\partial x_i^{\beta}}{\partial \beta} = x_i^{\beta}\log x_i, \qquad \frac{\partial(\alpha x_i^{\beta})}{\partial \beta} = \alpha x_i^{\beta}\log x_i.$$

Consequently,

$$\begin{aligned}\frac{\partial Q}{\partial \beta} &= 2\sum_{i=1}^{n}(y_i - \alpha x_i^{\beta})(-\alpha x_i^{\beta}\log x_i) \\ &= -2\alpha\sum_{i=1}^{n}(y_i - \alpha x_i^{\beta})x_i^{\beta}\log x_i.\end{aligned}$$

If $\alpha \neq 0$, the second normal equation reduces to

$$\sum_{i=1}^{n}(y_i - \alpha x_i^{\beta})x_i^{\beta}\log x_i = 0. \tag{3}$$

Substitute (2) into (3). We must solve

$$\sum_{i=1}^{n}\{y_i - \widehat{\alpha}(\beta)x_i^{\beta}\}x_i^{\beta}\log x_i = 0.$$

If $\widehat{\beta}$ is a suitable minimizing solution, then

$$\sum_{i=1}^{n}\{y_i - \widehat{\alpha}(\widehat{\beta})x_i^{\widehat{\beta}}\}x_i^{\widehat{\beta}}\log x_i = 0, \qquad \widehat{\alpha} = \widehat{\alpha}(\widehat{\beta}).$$

This kind of model is called a *partial linear model*: the model is linear in α for a fixed β. It is also sometimes called a separable non-linear regression model.

5 Properties and Geometry of LS estimators

Having obtained $\widehat{\alpha}$ and $\widehat{\beta}$, we may ask whether

$$\widehat{\alpha} \xrightarrow{P} \alpha, \qquad \widehat{\beta} \xrightarrow{P} \beta.$$

Although we mention in-probability convergence here, in many cases almost-sure convergence holds under appropriate conditions. We also wish to find asymptotic distributions and variances, in order to construct confidence intervals.

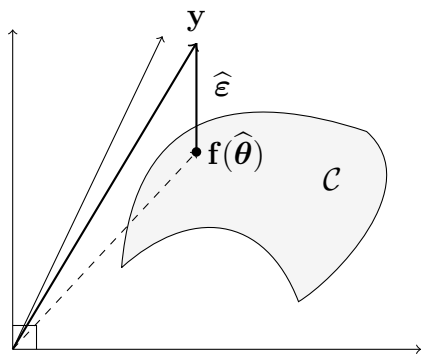
For the general model, use the notation

$$S(\boldsymbol{\theta}) = \sum_{i=1}^{n}\{y_i - f_i(\boldsymbol{\theta})\}^2 = \{\mathbf{y} - \mathbf{f}(\boldsymbol{\theta})\}^{\mathsf{T}}\{\mathbf{y} - \mathbf{f}(\boldsymbol{\theta})\}, \qquad \widehat{\boldsymbol{\theta}} \in \arg\min_{\boldsymbol{\theta}\in\Theta} S(\boldsymbol{\theta}).$$

Here S and Q denote the same sum of squares. Let $\boldsymbol{\theta}^* \in \Theta$ be the true parameter, and define the regression surface

$$\mathcal{C} = \{\mathbf{f}(\boldsymbol{\theta}) : \boldsymbol{\theta} \in \Theta\} \subseteq \mathbb{R}^n.$$

The LS fitted vector $\mathbf{f}(\widehat{\boldsymbol{\theta}})$ is a point on $\mathcal{C}$ closest to $\mathbf{y}$ in Euclidean distance.



A Schematic Diagram. The ambient space is $\mathbb{R}^n$. The oblique ray indicates a third direction in perspective.

5.1 Taylor approximation and distribution

At the true parameter,

$$y_i = f_i(\boldsymbol{\theta}^*) + \varepsilon_i, \qquad \mathbf{y} = \mathbf{f}(\boldsymbol{\theta}^*) + \boldsymbol{\varepsilon}.$$

We are assuming that $\widehat{\boldsymbol{\theta}}$ is close to $\boldsymbol{\theta}^*$ and then we want to study its distributional properties

5.1.1 First-order expansion

For each i, Taylor expansion around $\boldsymbol{\theta}^*$ gives

$$\begin{aligned} f_i(\boldsymbol{\theta}) &= f_i(\boldsymbol{\theta}^*) + \sum_{j=1}^{p} (\theta_j - \theta_j^*) \frac{\partial f_i(\boldsymbol{\theta}^*)}{\partial \theta_j} + r_i(\boldsymbol{\theta}) \\ &\approx f_i(\boldsymbol{\theta}^*) + \sum_{j=1}^{p} (\theta_j - \theta_j^*) \frac{\partial f_i(\boldsymbol{\theta}^*)}{\partial \theta_j}. \end{aligned}$$

Let $J_* = J(\boldsymbol{\theta}^*)$. In matrix notation,

$$\underbrace{\mathbf{f}(\boldsymbol{\theta})}_{n\times 1} \approx \underbrace{\mathbf{f}(\boldsymbol{\theta}^*)}_{n\times 1} + \underbrace{J_*}_{n\times p} \underbrace{(\boldsymbol{\theta} - \boldsymbol{\theta}^*)}_{p\times 1}.$$

Writing $\delta = \boldsymbol{\theta} - \boldsymbol{\theta}^*$,

$$\begin{aligned} S(\boldsymbol{\theta}) &= \{\mathbf{y} - \mathbf{f}(\boldsymbol{\theta})\}^\mathsf{T} \{\mathbf{y} - \mathbf{f}(\boldsymbol{\theta})\} \\ &\approx \{\mathbf{y} - \mathbf{f}(\boldsymbol{\theta}^*) - J_*\delta\}^\mathsf{T} \{\mathbf{y} - \mathbf{f}(\boldsymbol{\theta}^*) - J_*\delta\} \\ &= (\boldsymbol{\varepsilon} - J_*\delta)^\mathsf{T} (\boldsymbol{\varepsilon} - J_*\delta) \\ &= \boldsymbol{\varepsilon}^\mathsf{T}\boldsymbol{\varepsilon} - \boldsymbol{\varepsilon}^\mathsf{T} J_*\delta - \delta^\mathsf{T} J_*^\mathsf{T}\boldsymbol{\varepsilon} + \delta^\mathsf{T} J_*^\mathsf{T} J_*\delta \\ &= \boldsymbol{\varepsilon}^\mathsf{T}\boldsymbol{\varepsilon} - 2\delta^\mathsf{T} J_*^\mathsf{T}\boldsymbol{\varepsilon} + \delta^\mathsf{T} J_*^\mathsf{T} J_*\delta \\ &= L(\delta), \quad \text{(say)}. \end{aligned}$$

5.1.2 Minimizing the linearized criterion

Therefore

$$\frac{\partial L}{\partial \delta} = -2J_*^\mathsf{T}\boldsymbol{\varepsilon} + 2J_*^\mathsf{T} J_*\delta.$$

Setting the gradient to zero gives, successively,

$$J_*^\mathsf{T} J_*\widetilde{\delta} = J_*^\mathsf{T}\boldsymbol{\varepsilon}, \qquad \widetilde{\delta} = (J_*^\mathsf{T} J_*)^{-1} J_*^\mathsf{T}\boldsymbol{\varepsilon}.$$

The inverse exists when J_* has full column rank. This is an exact expression for the minimizer of the linearized criterion, but generally only an approximation for the original non-linear LS estimator. Hence

$$\widehat{\boldsymbol{\theta}} - \boldsymbol{\theta}^* \approx \widetilde{\delta} = (J_*^\mathsf{T} J_*)^{-1} J_*^\mathsf{T}\boldsymbol{\varepsilon}. \tag{4}$$

Remark 4 (Why $\widehat{\boldsymbol{\theta}} - \boldsymbol{\theta}^*$ need not equal $\widetilde{\delta}$). There are two different minimization problems. If we *define* $\widehat{\delta} := \widehat{\boldsymbol{\theta}} - \boldsymbol{\theta}^*$, this is an exact equality by definition. Changing variables in the original problem gives

$$\widehat{\delta} \in \underset{\delta:\, \boldsymbol{\theta}^* + \delta \in \Theta}{\arg\min} \left\| \mathbf{y} - \mathbf{f}(\boldsymbol{\theta}^* + \delta) \right\|^2.$$

But $\widetilde{\delta}$ was defined as the unrestricted minimizer of the *linearized* problem,

$$\widetilde{\delta} := \underset{\delta \in \mathbb{R}^p}{\arg\min} \, \|\boldsymbol{\varepsilon} - J_*\delta\|^2 = (J_*^\mathsf{T} J_*)^{-1} J_*^\mathsf{T}\boldsymbol{\varepsilon}.$$

5.2 Mean and variance–covariance matrix

Put $B = (J_*^\mathsf{T} J_*)^{-1} J_*^\mathsf{T}$, a $p \times n$ matrix. Since $\mathbb{E}\boldsymbol{\varepsilon} = 0$,

$$\mathbb{E}(\widetilde{\delta}) = B\mathbb{E}(\boldsymbol{\varepsilon}) = 0.$$

Since $\mathrm{Cov}(\boldsymbol{\varepsilon}) = \sigma^2 I_n$,

$$\begin{aligned}\mathrm{Cov}(\widetilde{\delta}) &= B\,\mathrm{Cov}(\boldsymbol{\varepsilon})B^\mathsf{T} \\ &= \sigma^2 (J_*^\mathsf{T} J_*)^{-1} J_*^\mathsf{T} J_* (J_*^\mathsf{T} J_*)^{-1} \\ &= \sigma^2 (J_*^\mathsf{T} J_*)^{-1}.\end{aligned}$$

Accordingly, to first order,

$$\mathbb{E}(\widehat{\boldsymbol{\theta}} - \boldsymbol{\theta}^*) \approx 0, \qquad \mathrm{Cov}(\widehat{\boldsymbol{\theta}}) \approx \sigma^2 (J_*^\mathsf{T} J_*)^{-1}.$$

Remark 5 (First-order unbiasedness). This does not establish exact unbiasedness of nonlinear LS. The zero mean derived above belongs exactly to $\widetilde{\delta}$, a Taylor remainder can introduce finite-sample bias in $\widehat{\boldsymbol{\theta}}$.

If $\varepsilon_i \overset{\text{i.i.d.}}{\sim} N(0, \sigma^2)$, a linear transformation of the Gaussian error vector gives

$$\widetilde{\delta} = B\boldsymbol{\varepsilon} \sim N_p\big(0, \sigma^2 (J_*^\mathsf{T} J_*)^{-1}\big)\,.$$

Thus the corresponding normal law for $\widehat{\boldsymbol{\theta}} - \boldsymbol{\theta}^*$ is a first-order approximation.

5.3 An Asymptotic Result

To state an actual asymptotic result, one needs consistency, local smoothness, identifiability, control of the Taylor remainder, and a suitable central limit theorem. If these conditions hold, and

$$\frac{1}{n} J_*^\mathsf{T} J_* \longrightarrow A > 0, \qquad \frac{1}{\sqrt{n}} J_*^\mathsf{T} \boldsymbol{\varepsilon} \xrightarrow{d} N_p(0, \sigma^2 A),$$

with the remainder in (4) of order $o_P(n^{-1/2})$, then

$$\begin{aligned}\sqrt{n}(\widehat{\boldsymbol{\theta}} - \boldsymbol{\theta}^*) &= \left(\frac{1}{n} J_*^\mathsf{T} J_*\right)^{-1} \frac{1}{\sqrt{n}} J_*^\mathsf{T} \boldsymbol{\varepsilon} + o_P(1) \\ &\xrightarrow{d} N_p(0, \sigma^2 A^{-1}).\end{aligned}$$

6 Linear-model analogy and the residual distribution

In the linear model $\mathbf{y} = X\beta + \boldsymbol{\varepsilon}$, under normal errors and full column rank,

$$\widehat{\beta} - \beta = (X^\mathsf{T} X)^{-1} X^\mathsf{T} \boldsymbol{\varepsilon} \sim N_p\big(0, \sigma^2 (X^\mathsf{T} X)^{-1}\big)\,.$$

In non-linear regression, $J(\boldsymbol{\theta}^*)$ plays the role of X in the local approximation. In practice one substitutes $J(\widehat{\boldsymbol{\theta}})$, since $\boldsymbol{\theta}^*$ is unknown.

Define

$$P_* = J_*(J_*^\mathsf{T} J_*)^{-1} J_*^\mathsf{T}.$$

The fitted response has the first-order expansion

$$\begin{aligned}\mathbf{f}(\widehat{\boldsymbol{\theta}}) &\approx \mathbf{f}(\boldsymbol{\theta}^*) + J_*(\widehat{\boldsymbol{\theta}} - \boldsymbol{\theta}^*) \\ &\approx \mathbf{f}(\boldsymbol{\theta}^*) + J_*(J_*^\mathsf{T} J_*)^{-1} J_*^\mathsf{T} \boldsymbol{\varepsilon} \\ &= \mathbf{f}(\boldsymbol{\theta}^*) + P_* \boldsymbol{\varepsilon}.\end{aligned}$$

Therefore

$$\begin{aligned}\widehat{\boldsymbol{\varepsilon}} = \mathbf{y} - \mathbf{f}(\widehat{\boldsymbol{\theta}}) &\approx \mathbf{y} - \mathbf{f}(\boldsymbol{\theta}^*) - P_* \boldsymbol{\varepsilon} \\ &= \boldsymbol{\varepsilon} - P_* \boldsymbol{\varepsilon} \\ &= (I_n - P_*)\boldsymbol{\varepsilon}.\end{aligned}$$

Since $P_*^\mathsf{T} = P_*$ and $P_*^2 = P_*$,

$$\begin{aligned}(I_n - P_*)^\mathsf{T}(I_n - P_*) &= I_n - 2P_* + P_*^2 \\ &= I_n - P_*.\end{aligned}$$

It follows that

$$\begin{aligned}S(\widehat{\boldsymbol{\theta}}) &= \widehat{\boldsymbol{\varepsilon}}^\mathsf{T} \widehat{\boldsymbol{\varepsilon}} \\ &\approx \boldsymbol{\varepsilon}^\mathsf{T}(I_n - P_*)^\mathsf{T}(I_n - P_*)\boldsymbol{\varepsilon} \\ &= \boldsymbol{\varepsilon}^\mathsf{T}(I_n - P_*)\boldsymbol{\varepsilon}.\end{aligned}$$

The symmetric idempotent matrix $I_n - P_*$ has rank $n - p$. Under Gaussian errors,

$$\frac{\boldsymbol{\varepsilon}^\mathsf{T}(I_n - P_*)\boldsymbol{\varepsilon}}{\sigma^2} \sim \chi^2_{n-p}. \tag{5}$$

For example, an orthogonal diagonalization of $I_n - P_*$ turns the quadratic form into a sum of squares of $n - p$ independent standard normal variables. Equation (5) is exact for this fixed-projection quadratic form, replacing its numerator by the actual non-linear residual sum of squares is an approximation.

7 Confidence sets

At the true parameter, the exact identity is

$$S(\boldsymbol{\theta}^*) = \{\mathbf{y} - \mathbf{f}(\boldsymbol{\theta}^*)\}^\mathsf{T}\{\mathbf{y} - \mathbf{f}(\boldsymbol{\theta}^*)\} = \boldsymbol{\varepsilon}^\mathsf{T}\boldsymbol{\varepsilon}.$$

Also,

$$S(\widehat{\boldsymbol{\theta}}) = \{\mathbf{f}(\boldsymbol{\theta}^*) + \boldsymbol{\varepsilon} - \mathbf{f}(\widehat{\boldsymbol{\theta}})\}^\mathsf{T}\{\mathbf{f}(\boldsymbol{\theta}^*) + \boldsymbol{\varepsilon} - \mathbf{f}(\widehat{\boldsymbol{\theta}})\}.$$

Subtracting and applying the linearized residual formula,

$$\begin{aligned} S(\boldsymbol{\theta}^*) - S(\widehat{\boldsymbol{\theta}}) &\approx \boldsymbol{\varepsilon}^\mathsf{T}\boldsymbol{\varepsilon} - \boldsymbol{\varepsilon}^\mathsf{T}(I_n - P_*)\boldsymbol{\varepsilon} \\ &= \boldsymbol{\varepsilon}^\mathsf{T}\{I_n - (I_n - P_*)\}\boldsymbol{\varepsilon} \\ &= \boldsymbol{\varepsilon}^\mathsf{T} P_* \boldsymbol{\varepsilon}. \end{aligned}$$

The two projection matrices satisfy

$$P_*(I_n - P_*) = P_* - P_*^2 = 0.$$

Under Gaussian errors, $P_*\boldsymbol{\varepsilon}$ and $(I_n - P_*)\boldsymbol{\varepsilon}$ are jointly normal and have cross-covariance

$$\operatorname{Cov}(P_*\boldsymbol{\varepsilon}, (I_n - P_*)\boldsymbol{\varepsilon}) = \sigma^2 P_*(I_n - P_*) = 0.$$

They are therefore independent, as are their squared norms. Consequently,

$$\frac{\boldsymbol{\varepsilon}^\mathsf{T} P_* \boldsymbol{\varepsilon}}{\sigma^2} \sim \chi^2_p, \qquad \frac{\boldsymbol{\varepsilon}^\mathsf{T}(I_n - P_*)\boldsymbol{\varepsilon}}{\sigma^2} \sim \chi^2_{n-p},$$

independently. The associated independence statements for the non-linear sums of squares are first-order approximations.

7.1 Linear-model confidence sets

Let γ denote the significance level, to avoid confusion with the power-law parameter α. Write $\chi^2_{p,1-\gamma}$ and $F_{p,n-p;1-\gamma}$ for lower-tail $(1-\gamma)$ quantiles. For the linear model define

$$W(\beta) = (\widehat{\beta} - \beta)^\mathsf{T} X^\mathsf{T} X (\widehat{\beta} - \beta), \quad R = (\mathbf{y} - X\widehat{\beta})^\mathsf{T}(\mathbf{y} - X\widehat{\beta}).$$

When σ^2 is known, an exact $(1-\gamma)$ confidence set is

$$\mathcal{A}_{1-\gamma} = \{\beta : W(\beta) \le \sigma^2 \chi^2_{p,1-\gamma}\}.$$

When σ^2 is unknown, $\widehat{\sigma}^2 = R/(n-p)$ and

$$\begin{aligned} \frac{W(\beta)/p}{R/(n-p)} &= \frac{W(\beta)}{R}\frac{n-p}{p} \\ &= \frac{W(\beta)}{p\widehat{\sigma}^2} \sim F_{p,n-p} \end{aligned}$$

at the true β. Hence

$$\mathcal{B}_{1-\gamma} = \left\{ \beta : \frac{W(\beta)}{R}\frac{n-p}{p} \le F_{p,n-p;1-\gamma} \right\}.$$

7.2 The non-linear analogue

From $\widetilde{\delta} = (J_*^\mathsf{T} J_*)^{-1} J_*^\mathsf{T}\boldsymbol{\varepsilon}$,

$$\begin{aligned} \widetilde{\delta}^\mathsf{T} J_*^\mathsf{T} J_* \widetilde{\delta} &= \boldsymbol{\varepsilon}^\mathsf{T} J_*(J_*^\mathsf{T} J_*)^{-1}(J_*^\mathsf{T} J_*)(J_*^\mathsf{T} J_*)^{-1} J_*^\mathsf{T}\boldsymbol{\varepsilon} \\ &= \boldsymbol{\varepsilon}^\mathsf{T} J_*(J_*^\mathsf{T} J_*)^{-1} J_*^\mathsf{T}\boldsymbol{\varepsilon} \\ &= \boldsymbol{\varepsilon}^\mathsf{T} P_* \boldsymbol{\varepsilon}. \end{aligned}$$

Thus, evaluated at the true parameter,

$$\frac{(\widehat{\boldsymbol{\theta}} - \boldsymbol{\theta}^*)^\mathsf{T} J_*^\mathsf{T} J_* (\widehat{\boldsymbol{\theta}} - \boldsymbol{\theta}^*)}{\sigma^2} \overset{\text{approx.}}{\sim} \chi^2_p.$$

Since $\boldsymbol{\theta}^*$ is unknown, we substitute

$$\widehat{J} = J(\widehat{\boldsymbol{\theta}}), \qquad \widehat{\sigma}^2 = \frac{S(\widehat{\boldsymbol{\theta}})}{n-p}, \qquad W_N(\boldsymbol{\theta}) = (\widehat{\boldsymbol{\theta}} - \boldsymbol{\theta})^\mathsf{T} \widehat{J}^\mathsf{T} \widehat{J} (\widehat{\boldsymbol{\theta}} - \boldsymbol{\theta}).$$

An approximate confidence ellipsoid for known variance is

$$\left\{ \boldsymbol{\theta} \in \Theta : W_N(\boldsymbol{\theta}) \le \sigma^2 \chi^2_{p,1-\gamma} \right\}.$$

For unknown variance, the local Gaussian linear-model approximation gives

$$\left\{ \boldsymbol{\theta} \in \Theta : \frac{W_N(\boldsymbol{\theta})/p}{S(\widehat{\boldsymbol{\theta}})/(n-p)} \le F_{p,n-p;1-\gamma} \right\}.$$

Neither is generally an exact finite-sample confidence region.